

error-prone due to boredom and distractions. Machines, however, do not mind going through the same routine, over and over, and they perform routine, repetitive tasks much faster and more efficiently than people do.

But that doesn't mean they always get it right. Even with AI, it is necessary to keep an eye on the work to check whether the algorithms are still working within the desired parameters. AI and ML without human interference might drift from the set path. But working in partnership with AI, researchers are relieved of the burden of menial work.

The impact of AI and ML

The past five years have seen a tremendous rise in the use of AI and ML technologies for enterprises. Most applications can be attributed to advancements in computing power and the evolution of paradigms like distributed computing, Big Data and cloud computing. Early commercial applications of ML were pioneered by technology giants like Google, Amazon and Facebook. These businesses managed to build a store of valuable behavioural data from millions of users. In order to effectively collect, cleanse, organise and analyse their consumer data, these companies built scalable Big Data frameworks and applications which were then open sourced to the world. This helped these frameworks to improve fast and allowed businesses to derive more value from their data. Organisations are already beginning to use AI to bolster

cyber security and offer more protection against sophisticated hackers. AI helps by automating complex processes for detecting attacks and reacting to breaches.

Data deception technology products can automatically detect, analyse and defend systems against advanced attacks by proactively detecting attackers. So, when one combines security personnel with adaptive technology that continues to change and become smarter over time, it provides a competitive edge to defenders that has till now been absent from most cyber security technologies.

On the other hand, AI can open up vulnerabilities as well. This happens particularly when AI depends on interfaces within and across organisations that create access opportunities by *bad actors* or disreputable agents. And attackers are beginning to deploy AI, too, in ways that give computer programs the ability to make decisions that benefit the attackers. This means that these programs will gradually develop automated hacks that are able to study and learn about the systems they target, and identify vulnerabilities.

How AI can benefit IoT

Fast detection of attacks and limiting their spread is what only AI and its algorithms as well as datasets can do. According to Steve Grobman, chief technology officer for McAfee, AI will be the cornerstone of tomorrow's cyber defence.

User and entity behaviour analytics (UEBA), which is what SIEM (security information and event management) solutions are based on, is another major IT investment trend. This technique uses machine learning capabilities to analyse behaviour logs and network traffic in real-time, and respond appropriately in the event of an attack. This is done by getting the user to log in again, blocking an attack or assessing risk levels and alerting the company's cyber security managers so that they can take the necessary action.

Artificial intelligence and machine learning have already gained a foothold in cyber security, and will only become stronger as the two fields are a perfect fit. The volume of new data coming in every day is too much for cost-effective human processing and machines are less error-prone, if trained properly. There will be some kinks to work out, as AI and ML are still very much in the development phases. The expectation is that the widespread use of AI and ML will reduce the quantity of the mundane work we humans do, but we will then have to step up and take on more challenging tasks. **END** 🐧

Reference

- [1] <https://blog.econocom.com/en/blog/artificial-intelligence-a-real-boon-for-iot-security/>

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